STAT/MA 41600
In-Class Problem Set #37: November 13, 2026
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Problem Set 37 Answers

In this problem set, Z always represents a standard normal random variable.

In questions 1, 3, 4, and 5, we use continuity correction since the random variables in those questions are integer valued.

1. Let X denote the number of raindrops that fall during the fixed period of time. We compute $P(X \geq 295) = P(X \geq 294.5) = P((X - 300)/\sqrt{300} \geq (294.5 - 300)/\sqrt{300}) \approx P(Z \geq -0.32) = P(Z \leq 0.32) = 0.6255$.

2ab. The random variable X is a Gamma random variable with $r = 400$ and $\lambda = 1/2$.

2c. Yes, the distribution of X is approximately Normal, with $E(X) = r/\lambda = (400)(2) = 800$ and $\text{Var}(X) = r/\lambda^2 = (400)(4) = 1600$.

2d. We compute $P(X > 815) = P((X - 800)/\sqrt{1600} > (815 - 800)/\sqrt{1600}) \approx P(Z > 0.375) = 1 - P(Z \leq 0.375) = 1 - 0.6480 = 0.3520$.

3. There are always 500 (regular) blue bears in the boxes, automatically, without taking the bonus bears into account. So there are *strictly more than* 580 blue bears altogether if and only if there are *strictly more than* 80 bonus blue bears.

The number of bonus bears that are blue is a Binomial random variable X with parameters $n = 500$ and $p = 1/6$.

We compute $P(X > 80) = P(X > 80.5) = P\Big((X - (500)(1/6))/\sqrt{(500)(1/6)(5/6)} > (80.5 - (500)(1/6))/\sqrt{(500)(1/6)(5/6)}\Big) \approx P(Z > -0.34) = P(Z < 0.34) = 0.6331$.

4. We know that $X = X_1 + \cdots + X_{50}$ where X_i is the number of blue results that occur between the $(i-1)$st red roll and the ith red roll.

These X_is are independent. The easiest way to get the distribution of the X_is is to ignore the yellow outcomes, i.e., to view the dice as having 1 red side and 2 blue sides. Each X_i is a Geometric number-of-losses random variables, with $E(X_i) = \frac{1}{1/3} - 1 = 2$ and $\text{Var}(X_i) = (2/3)/(1/3)^2 = 6$. It follows that $E(X) = (50)(2) = 100$ and $\text{Var}(X) = (50)(6) = 300$.

We compute $P(97 \leq X \leq 103) = P(96.5 \leq X \leq 103.5) = P((96.5 - 100)/\sqrt{300} \leq (X - 100)/\sqrt{300} \leq (103.5 - 100)/\sqrt{300} \approx P(-0.20 \leq Z \leq 0.20) = P(Z \leq 0.20) - P(Z < -0.20) = P(Z \leq 0.20) - P(Z > 0.20) = P(Z \leq 0.20) - (1 - P(Z \leq 0.20)) = 2P(Z \leq 0.20) - 1 = (2)(0.5793) - 1 = 0.1586$.

5. From Question 4, we know that X is approximately Normal with $E(X) = (50)(2) = 100$ and $\text{Var}(X) = (50)(6) = 300$. Also, Poisson random variables with large parameter are approximately Normal, so Y is approximately Normal with $E(Y) = 100$ and $\text{Var}(Y) = 100$. Since X and Y are independent, it follows that $X - Y$ is approximately Normal with $E(X - Y) = 100 - 100 = 0$ and $\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) = 300 + 100 = 400$. Thus $P(|X - Y| \leq 10) = P(-10 \leq X - Y \leq 10) = P(-10.5 \leq X - Y \leq 10.5) = P((-10.5 - 0)/\sqrt{400} \leq (X - Y - 0)/\sqrt{400} \leq (10.5 - 0)/\sqrt{400}) \approx P(-0.53 \leq Z \leq 0.53) = P(Z \leq 0.53) - P(Z < -0.53) = P(Z \leq 0.53) - P(Z > 0.53) = P(Z \leq 0.53) - (1 - P(Z \leq 0.53)) = 2P(Z \leq 0.53) - 1 = (2)(0.7019) - 1 = 0.4038$.